

6.1

$$P(\text{cancer}) = 0.008$$

$$P(\neg \text{cancer}) = 0.992$$

$$P(+|\text{cancer}) = 0.98$$

$$P(-|\text{cancer}) = 0.02$$

$$P(+|\neg \text{cancer}) = 0.03$$

$$P(-|\neg \text{cancer}) = 0.97$$

所以：

$$\begin{aligned} P(\text{cancer}|++) &= \frac{P(++|\text{cancer})P(\text{cancer})}{P(++)} \\ &= \frac{P(++|\text{cancer})P(\text{cancer})}{P(++|\text{cancer})P(\text{cancer}) + P(++|\neg \text{cancer})P(\neg \text{cancer})} \\ &= \frac{0.98 \times 0.98 \times 0.008}{0.98 \times 0.98 \times 0.008 + 0.03 \times 0.03 \times 0.992} \\ &= 0.0076832 \end{aligned}$$

$$\begin{aligned} P(\neg \text{cancer}|++) &= \frac{P(++|\neg \text{cancer})P(\neg \text{cancer})}{P(++|\text{cancer})P(\text{cancer}) + P(++|\neg \text{cancer})P(\neg \text{cancer})} \\ &= \frac{0.03 \times 0.03 \times 0.992}{0.98 \times 0.98 \times 0.008 + 0.03 \times 0.03 \times 0.992} \\ &= 0.0008928 \end{aligned}$$

归一化结果：

$$\begin{aligned} P(\text{cancer}|++) &= \frac{P(++|\text{cancer})P(\text{cancer})}{P(++|\text{cancer})P(\text{cancer}) + P(++|\neg \text{cancer})P(\neg \text{cancer})} \\ &= 0.8959 \end{aligned}$$

$$\begin{aligned} P(\neg \text{cancer}|++) &= \frac{P(++|\neg \text{cancer})P(\neg \text{cancer})}{P(++|\text{cancer})P(\text{cancer}) + P(++|\neg \text{cancer})P(\neg \text{cancer})} \\ &= 0.1041 \end{aligned}$$

6.3

根据贝叶斯公式，后验概率可以表示为：

$$P(\text{cancer}|+) = \frac{P(+|\text{cancer})P(\text{cancer})}{P(+)}$$

全概率公式：

公式中的分母 $P(+)$ 是测试结果为阳性的全概率。根据全概率公式，可以将其展开为所有互斥情况的概率之和：

$$\begin{aligned} P(+) &= P(+, \text{cancer}) + P(+, \neg\text{cancer}) \\ &= P(+|\text{cancer})P(\text{cancer}) + P(+|\neg\text{cancer})P(\neg\text{cancer}) \end{aligned}$$

将 $P(+)$ 的展开式代入第2步的贝叶斯公式中，得到：

$$P(\text{cancer}|+) = \frac{P(+|\text{cancer})P(\text{cancer})}{P(+|\text{cancer})P(\text{cancer}) + P(+|\neg\text{cancer})P(\neg\text{cancer})}$$

观察此公式可以发现：分子是 $P(+|\text{cancer}) P(\text{cancer})$ 。分母是 $P(+|\text{cancer}) P(\text{cancer})$ 与 $P(+|\neg\text{cancer}) P(\neg\text{cancer})$ 的和。这与将 $P(+|\text{cancer}) P(\text{cancer})$ 和 $P(+|\neg\text{cancer}) P(\neg\text{cancer})$ 归一化的操作完全吻合。因此，该归一化方法是正确的，它本质上是贝叶斯公式和全概率公式的一种直接应用。